

Autonomous Aerospace Monitoring Assistant:

An Integrative AI-Based Anomaly Triage
Framework



01.

Industry context & problem definition

- **Aerospace operations rely on continuous telemetry + visual sensor streams (multimodal).**
- Anomalies can be benign noise, environmental effects, or critical malfunctions.

— The core challenge:

Responsible triage under uncertainty (not detection alone).

— Why AI:

High-volume, high-dimensional data; nonlinear patterns; probabilistic reasoning.

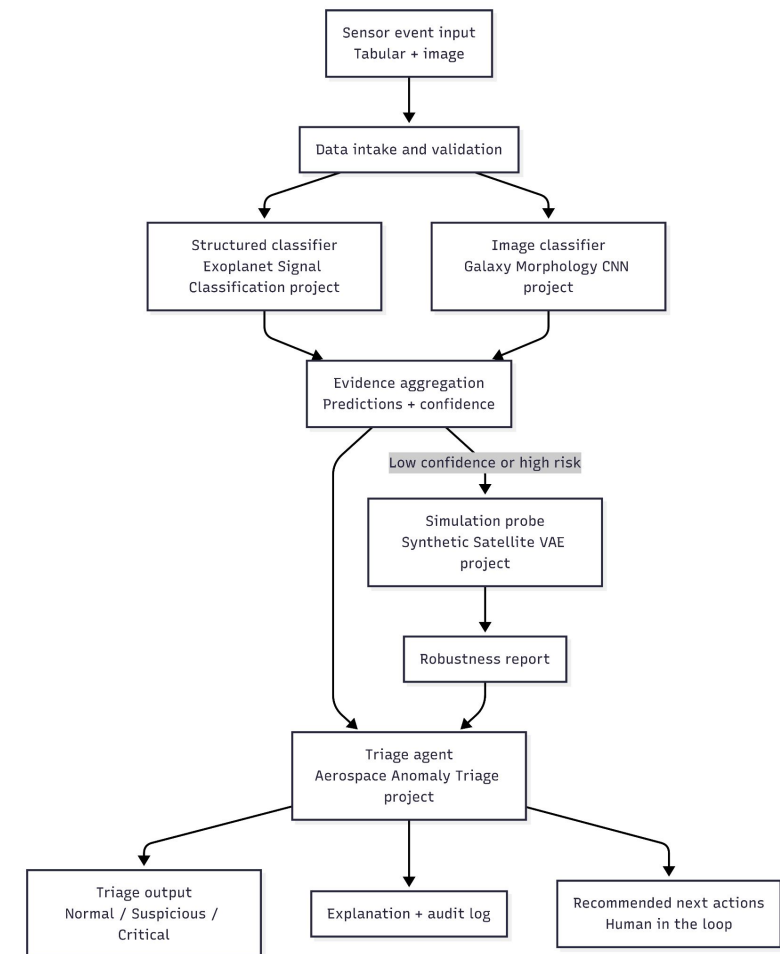
— Constraint:

Safety-critical setting → transparency, auditability, and human oversight.

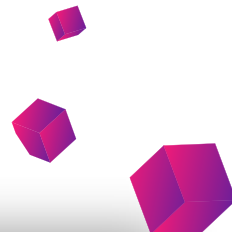
02.

Integrated AI system overview

Figure: Modular workflow. Classifiers → evidence aggregation → conditional robustness probe → triage + audit log + human-in-the-loop actions.



03. Integration of prior capstone projects



Prior capstone project	What it contributed	Where it appears in the system
Exoplanet Signal Classification (ML)	Structured/tabular classifier pattern + confidence	Structured classifier module + evidence fusion
Galaxy Morphology CNN (DL)	Image perception workflow + confidence	Image classifier module + multimodal corroboration
Synthetic Satellite VAE (GenAI)	Robustness probing via plausible perturbations	Conditional simulation/probe + robustness report
Aerospace Anomaly Triage (Agentic)	Bounded decision logic + safeguards	Triage agent + explanations + recommended next actions

Integration enables multimodal corroboration, robustness testing under uncertainty, and bounded decision logic — improving safety, auditability, and reliability compared to a single-model approach.

04. Key technical decisions & tradeoffs

01

Modular architecture
(auditability, replaceable modules) vs end-to-end optimization.

02

Evidence aggregation:
combine labels + confidence; detect modality disagreement.

03

Risk-aware routing:
low confidence / disagreement / data-quality flag → escalation.

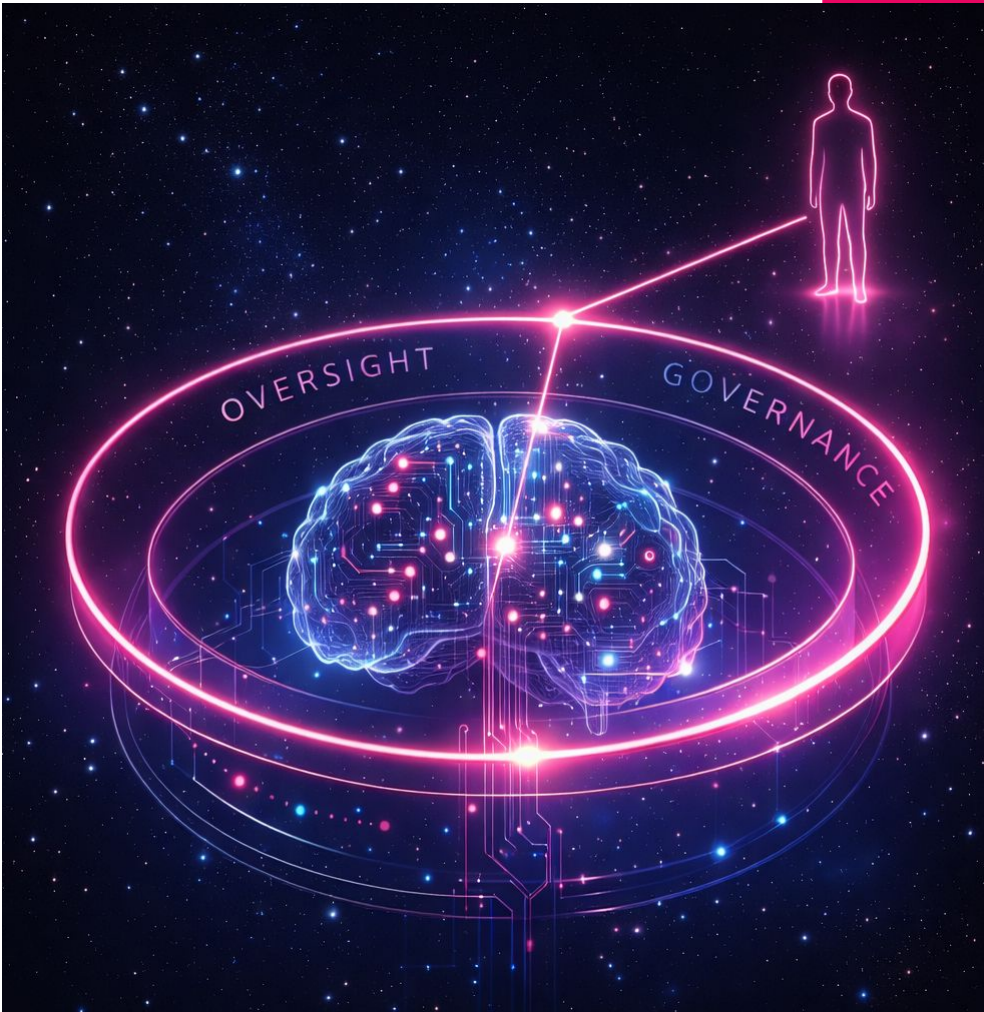
04

Conditional robustness probe:
run only when uncertainty or risk is high (cost-aware).

05

Hard safety override:
data integrity issues force human review (fail-safe).

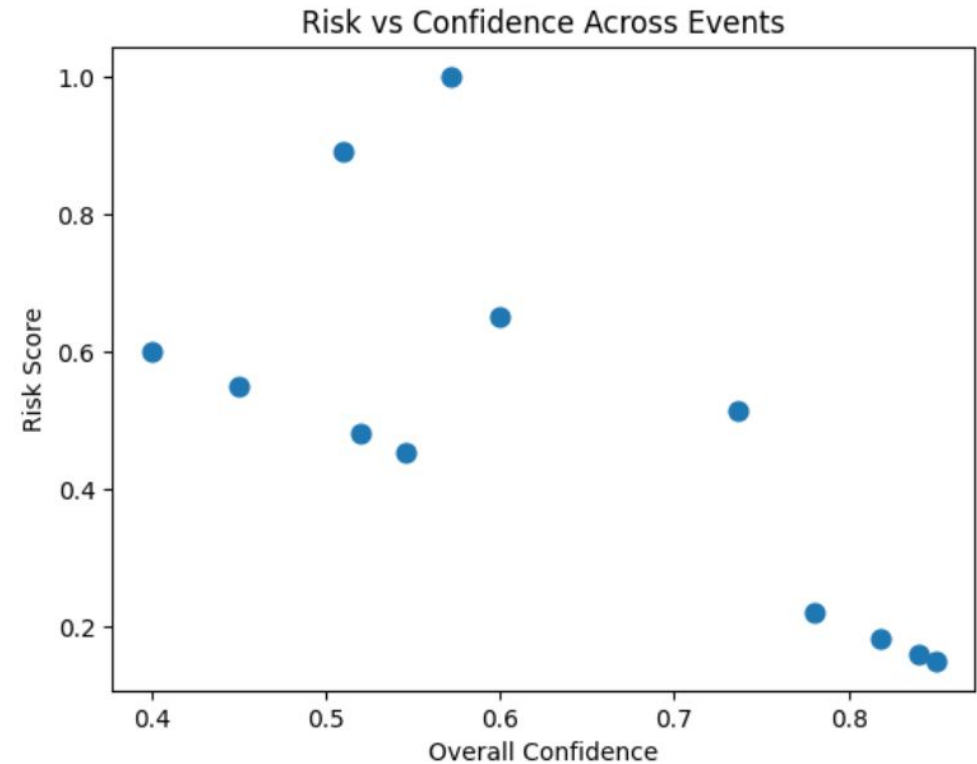
05. Ethical considerations & safeguards (system-specific)



- **Automation bias:** surface confidence + explanations; uncertainty triggers escalation.
- **Accountability:** structured audit log records evidence, thresholds, and routing decisions.
- **Transparency:** intermediate outputs preserved for post hoc review and governance.
- **Safety:** conservative escalation + human-in-the-loop for high-risk cases.
- **Bounded autonomy:** decision support only—no autonomous mission-critical actions.

06.

Evaluation: risk–confidence relationship



Lower confidence generally corresponds to higher operational risk; routing escalates accordingly.

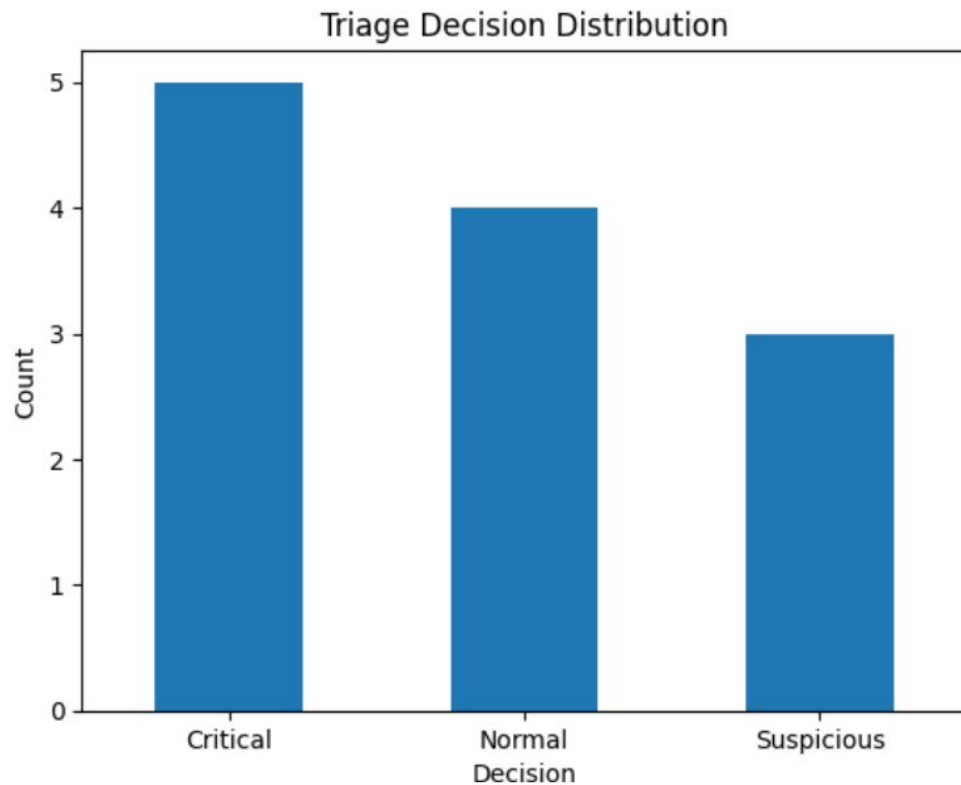
07. Evaluation: Conditional Robustness Probe Routing



The robustness probe activates selectively under low-confidence or higher-risk conditions, demonstrating conditional escalation rather than constant simulation.

08.

Evaluation: system behavior (scenario suite)



Observed distribution of Normal / Suspicious / Critical outcomes in the scenario suite.

09. Limitations, risks, and what would change in production

Prototype emphasizes orchestration and defensible routing—not production-grade model training.

01

Thresholds require calibration to mission context (cost of false negatives vs alert fatigue).

03

Correlated model error remains a systemic risk → diversify data and stress-test edge cases.

05

02

Robustness probe is a simplified stand-in; production would use validated generative/physics-informed tests.

04

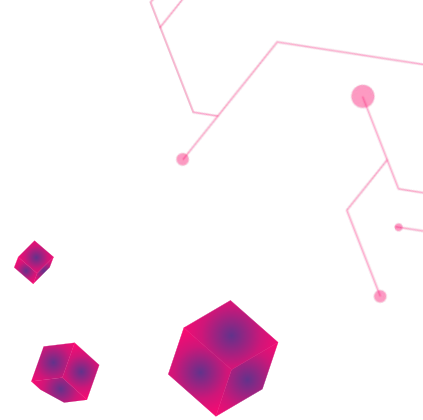
Distribution shift (sensor drift, environment changes) demands monitoring and periodic validation.

10.

Professional Readiness

Demonstrates readiness for applied AI system design in safety-critical domains

- **System-level architecture:** multimodal integration, routing logic, safeguards.
- **Cross-paradigm fluency:** ML + DL + GenAI + agentic decision logic.
- **Analytical reasoning:** risk modeling, uncertainty-aware routing.
- **Responsible AI design:** governance, auditability, human oversight



11.

Next Steps & Production Evolution

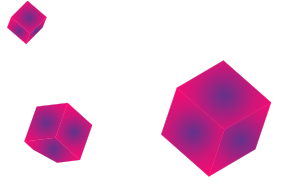
Path toward production deployment

- Integrate production-trained aerospace models
- Calibrate thresholds to mission cost functions
- Implement dataset drift monitoring
- Expand scenario validation and stress testing

From prototype orchestration → operational AI system

12. Autonomous Aerospace Monitoring Assistant

Defensible AI for Safety-Critical Decision Support



- **Structured reasoning**
- **Bounded autonomy**
- **Human-centered oversight**

Thank you.

